

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 6

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Topic

Rotating-Frame Corrections

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Rotating-Frame Corrections. State all assumptions and conventions.

$$g_{00}^{rot} = g_{00} + \mu_1 \Omega^2 r^2 / c^2$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$g_{\varphi\varphi}^{read} = g_{\varphi\varphi} (1 + \mu_2 \Omega^2 r^2 / c^2)$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$\mu_1 = 1, \mu_2 = -1/2$$

Show enough intermediate work that another student can reproduce the calculation.